

Abstract

Space debris in Low Earth Orbit poses an increasing collision risk, necessitating Active Debris Removal (ADR). Accurate orbit prediction is critical for ADR operations, yet standard Two-Line Element (TLE) sets achieve only kilometre-level accuracy, insufficient for rendezvous. This study evaluates Space Debris Laser Ranging (SDLR)-based orbit determination (OD) as a means of improving debris position predictions, applied to two LEO objects with contrasting eccentricities ($e = 0.005$ and $e = 0.069$). Using single-station SDLR measurements and batch least-squares estimation, two configurations are compared: fixed and estimated drag coefficient. SDLR-based OD improved orbit predictions by $2.6\text{--}617\times$ over TLE across short-term and medium-term intervals. However, drag coefficient estimation, while improving orbit fit, caused severe prediction degradation for the eccentric object at seven days due to atmospheric variability. Results suggest eccentricity governs drag estimation viability, with fixed-drag models providing more robust medium-term predictions. Operational recommendations for the Autonomous Removal and Maintenance Architecture (ARMA) are proposed.

Nomenclature

Acronyms

ADR	Active Debris Removal
ARMA	Autonomous Removal and Maintenance Architecture
BLS	Batch Least Squares
CRD	Consolidated Laser Ranging Data
ESA	European Space Agency
FRD	Full Rate Data
GCRF	Geocentric Celestial Reference Frame
ILRS	International Laser Ranging Service
ITRF	International Terrestrial Reference Frame
LEO	Low Earth Orbit
LVLH	Local Vertical Local Horizontal
NOAA	National Oceanic and Atmospheric Administration
NORAD	North American Aerospace Defense Command
NPT	Normal Point
OD	Orbit Determination
RAAN	Right Ascension of the Ascending Node
RMS	Root Mean Square
SDLR	Space Debris Laser Ranging
SGP4	Simplified General Perturbations 4
SLR	Satellite Laser Ranging
SRP	Solar Radiation Pressure
SSO	Sun-Synchronous Orbit
TLE	Two-Line Element

Greek Symbols

Δt	Round-trip travel time (s)
δ	Range measurement (m)
$\Delta\delta$	Range residual (m)
$\mu_{\oplus}$	Earth gravitational parameter ($\text{km}^3 \text{s}^{-2}$)
σ	Measurement uncertainty (m)
τ	Time of flight (s)

Scalar Symbols

ν	True anomaly (deg)
Ω	Right ascension of the ascending node (deg)
ω	Argument of perigee (deg)
a	Semi-major axis (km)
c	Speed of light (km s^{-1})
C_D	Drag coefficient (–)
e	Eccentricity (–)
i	Inclination (deg)
$J(\mathbf{x})$	Least-squares cost function (m^2)
N	Number of observations (–)
t	Time (s)
w_i	Observation weight (m^{-2})

Matrices and Vectors

$\mathbf{F}$	Force vector (N)
$\mathbf{H}$	Measurement Jacobian matrix $\partial\boldsymbol{\delta}/\partial\mathbf{x}$
$\mathbf{P}$	Solution covariance matrix
$\mathbf{r}$	Position vector (km)
$\mathbf{v}$	Velocity vector (km s^{-1})
$\mathbf{x}$	State vector $[\mathbf{r}, \mathbf{v}]^T$
$\mathbf{x}^*$	Optimal (best-fit) state vector
$\mathbf{x}_0$	Initial state vector (from TLE)
$\mathbf{x}_{aug}$	Augmented state vector $[\mathbf{r}, \mathbf{v}, C_D]^T$

Subscripts and Superscripts

$(\cdot)^*$	Optimal value
$(\cdot)_0$	Initial value
$(\cdot)_\oplus$	Earth
$(\cdot)_{aug}$	Augmented state
$(\cdot)_{comp}$	Computed measurement
$(\cdot)_i$	At each training measurement
$(\cdot)_{obs}$	Observed measurement

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Chapter 1

Introduction and Background

1.1 Rationale

1.1.1 The Space Debris Problem and Active Debris Removal



APPROXIMATELY 40,000 debris objects are currently tracked in Earth orbit, with hundreds of thousands of smaller fragments remaining untracked¹ (Figure 1.1). The problem is most severe in Low Earth Orbit (LEO), where in the European Space Agency's (ESA) Space Environment Report 2025, LEO now contains a comparable number of debris objects and active satellites.² Collisions amplify this risk through cascading fragmentation, a phenomenon known as Kessler syndrome, making passive mitigation alone insufficient.³ Active Debris Removal (ADR) is therefore required to preserve the long-term usability of the orbital environment.¹ It must contend with uncooperative debris targets, characterised by irregular shape, tumbling motion, and lack of control or communication.⁴ ADR has been demonstrated in orbit,⁵ establishing a baseline of proven technology and methods.

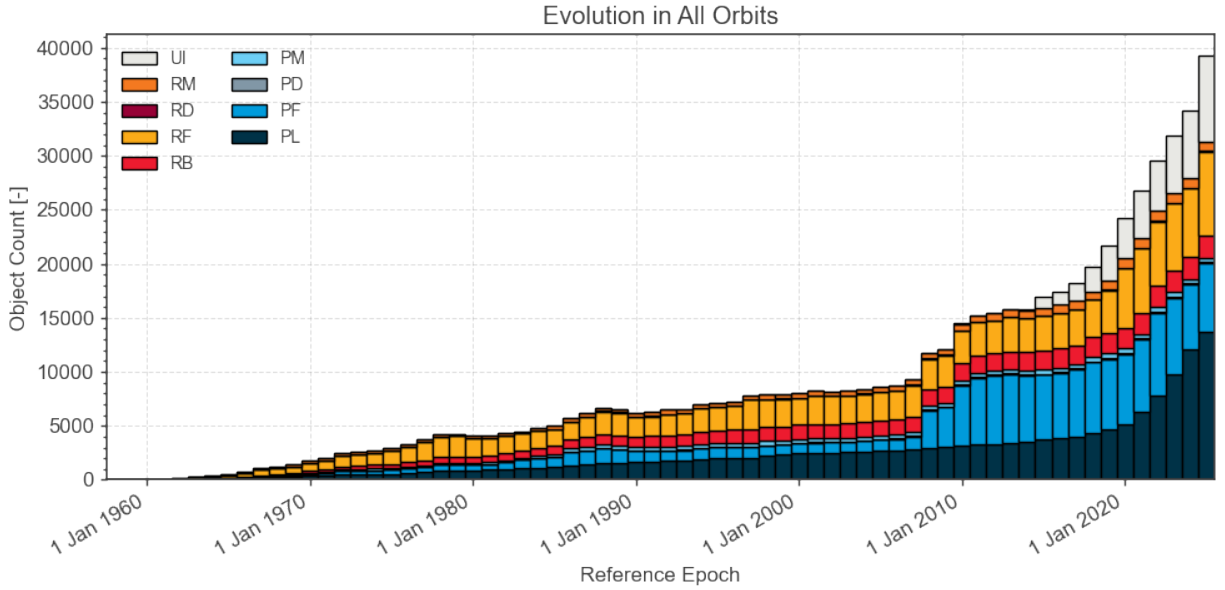


FIGURE 1.1. Excerpt from the European Space Agency’s 2025 Space Environment Report:² Number of tracked objects in Earth orbit over time. UI = Unidentified, RM = Rocket mission related object, RD = Rocket debris, RF = Rocket fragmentation debris, RB = Rocket body, PM = Payload mission related object, PD = Payload debris, PF = Payload fragmentation debris, PL = Payload

1.1.2 Proposed Solution

Within this context, the Autonomous Removal and Maintenance Architecture (ARMA) is proposed: a mothership deploys multiple small satellites to rendezvous with and deorbit debris in Sun-synchronous LEO. This approach builds on emerging small-satellite ADR concepts⁶ and aims to maximise flexibility, scalability, and efficiency. This report focuses on accurate debris orbit determination and prediction, underpinning target selection, approach and rendezvous planning. The literature review is integrated across Chapters 1 and 2.

1.2 Analysis background

ADR requires target position knowledge to within a few to tens of metres, achieved through combined ground-based and onboard tracking (requirement derived from ARMA). Uncooperative debris is difficult to track due to the absence of transponders and retroreflectors, reducing the effectiveness of conventional observation techniques. Their key physical parameters, including the area-to-mass ratio, are often poorly defined, which introduces uncertainty in modelling their interactions with the orbital environment, particularly atmospheric drag effects characterised by the drag coefficient C_D . Eccentricity further complicates this modelling by introducing altitude variation and variable atmospheric conditions (Figure 1.2).

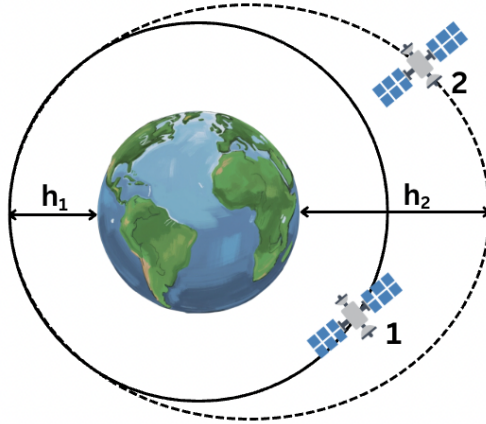


FIGURE 1.2. Description of orbital eccentricity. Satellite 1 is in a circular orbit, experiencing near-constant altitude h_1 , while satellite 2 is in a more eccentric orbit and experiences altitude variations from h_1 to h_2 .

Most debris objects are catalogued by the United States Space Force as Two-Line Element (TLE) sets. These datasets describe an object's orbit at a given epoch (point in time) and are propagated to estimate future position. Despite their widespread use in mission planning,⁷ TLE-based predictions for LEO debris only achieve accuracies on the order of 10^2 – 10^3 m,⁸ with errors growing rapidly with time since epoch. This accuracy is insufficient for ADR operations⁹ and higher-fidelity orbit prediction techniques are required.

One approach is to use predictions generated by an Orbit Determination (OD) process. OD estimates an object's orbit by refining an initial estimate by fitting it to a set of observations.¹⁰ The resulting solution represents the trajectory that best fits the observations and can be propagated to generate improved predictions. This study investigates improving space debris orbit predictions using OD based on laser ranging measurements.

1.3 Space Debris Laser Ranging as an OD Technique

1.3.1 Satellite Laser Ranging

Satellite Laser Ranging (SLR) determines the instantaneous distance (range) between a ground station and a satellite by measuring the round-trip travel time of laser pulses reflected from the satellite¹¹ (Figure 1.3).

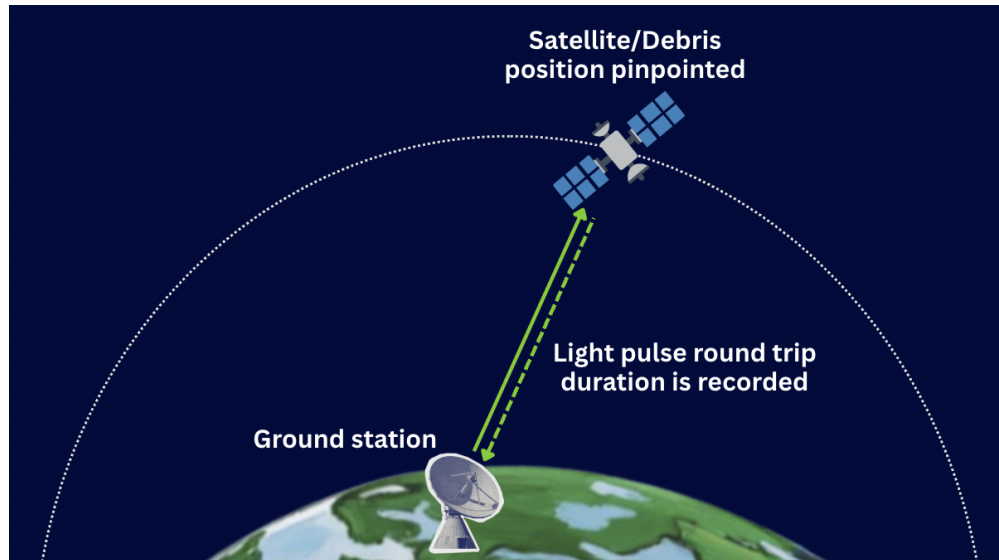


FIGURE 1.3. Conceptual illustration of the satellite laser ranging measurement principle.

Modern SLR systems are capable of measuring the pulse travel time with extreme precision, enabling range measurements with millimetre-level accuracy for satellites equipped with retroreflectors.¹¹

1.3.2 Space Debris Laser Ranging

Space Debris Laser Ranging (SDLR) extends SLR to uncooperative targets. Without retroreflectors, diffuse reflections produce weaker, more variable return signals, requiring higher-power lasers and more sensitive detectors.¹² Early demonstrations achieved decimetre-level precision, with modern systems reaching sub-decimetre accuracy at higher altitudes.¹³ Studies have shown that single-track SDLR observations can reduce OD uncertainty by 20–40% compared to TLE-based solutions,¹⁴ with advances such as daylight operation and multi-static tracking further improving availability and accuracy^{15–17}

In this study, OD is performed using observations from the SDLR-enabled station Izaña-2 (shown in Figure 1.4), provided by DiGOS Potsdam GmbH and ESA.

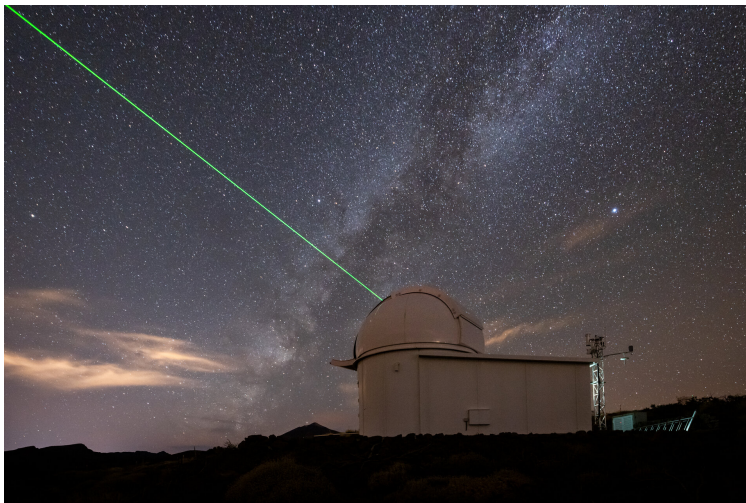


FIGURE 1.4. Picture of the SDLR-enabled station Izaña-2.

1.4 Research Gaps, Objectives, and Report Structure

Most SDLR OD studies focus on estimation accuracy rather than prediction degradation over operational timescales. Lejba *et al.*¹⁴ and Bamann & Hugentobler¹⁶ represent the closest existing work, though no study directly compares SDLR OD performance across different

eccentricity regimes, nor quantifies the implications of treating C_D as a variable parameter for prediction. Three research questions address these gaps:

- RQ1: What OD accuracy is achievable for LEO debris using single-station SDLR measurements, compared to TLE propagation?
- RQ2: Does estimating C_D alongside the orbital state improve OD and prediction accuracy, and does this depend on orbital eccentricity?
- RQ3: How does SDLR-based OD accuracy degrade over time as it is propagated, and what are the operational implications for ARMA?

The corresponding objectives are to: (1) implement SDLR-based OD for two debris objects with contrasting eccentricities; (2) compare baseline and C_D estimation configurations; and (3) assess short-term (2-day) and medium-term (7-day) prediction (OD propagation) accuracy.

This work adheres to the Royal Aeronautical Society’s Code of Conduct; data was provided under formal agreement and raises no personal data concerns. The dual-use nature of high-precision laser tracking is acknowledged, where the same infrastructure supporting debris removal could be applied to tracking operational or military satellites.

Active debris removal serves a clear societal interest: the orbital environment underpins critical global infrastructure including navigation, communications, weather forecasting, and climate monitoring. This work contributes to the technical foundation required to preserve it as a shared resource for future generations.

Chapter 2 presents theory, Chapter 3 methodology, Chapter 4 results, Chapter 5 discussion, and Chapter 6 conclusions and future work.

Chapter 2

Theory

2.1 Theory for Orbit Determination

2.1.1 Orbital Mechanics

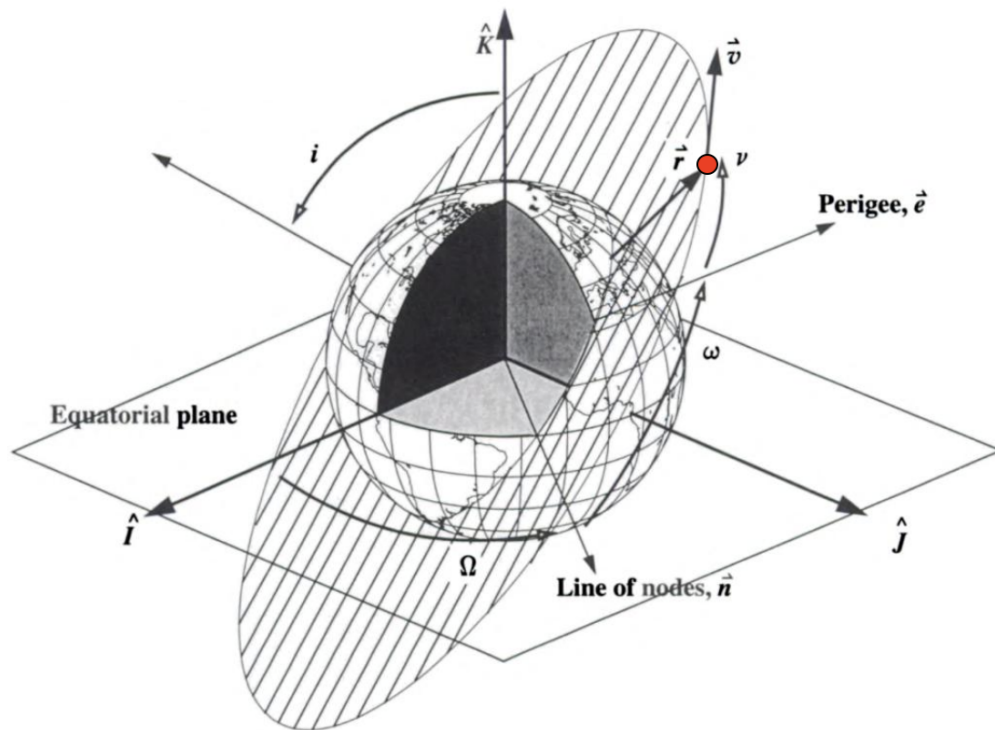


FIGURE 2.1. The six classical orbital elements with $\mathbf{r}$ and $\mathbf{v}$ shown. Taken from *Fundamentals of Astrodynamics and Applications*, D.A. Vallado¹⁸

The instantaneous motion of an orbiting debris object is described by a Cartesian state vector (Eq. 2.1) defined in a geocentric inertial reference frame (GCRF), comprising Earth-debris position $\mathbf{r}$ and velocity $\mathbf{v}$ components (Figure 2.1, the x , y , z directions correspond respectively to the unit vectors $\hat{I}$, $\hat{J}$, $\hat{K}$ along the axes).

$$\mathbf{x} = \begin{bmatrix} \mathbf{r} \\ \mathbf{v} \end{bmatrix} = \begin{bmatrix} x & y & z & \dot{x} & \dot{y} & \dot{z} \end{bmatrix}^T \quad (2.1)$$

The wider orbit can be physically interpreted using the six classical orbital elements (semi-major axis a , eccentricity e , inclination i , right ascension of the ascending node Ω , argument of perigee ω , and true anomaly ν) which describe its geometry¹⁸ (Figure 2.1).

Eccentricity governs altitude variation and therefore atmospheric drag, making it central to this analysis.

2.1.2 Orbital Perturbations

Classical orbital elements assume ideal motion influenced only by Earth’s central gravitational field. In reality, additional forces (perturbations) act on debris that must be modelled for accurate orbit determination and prediction.

In this study, the perturbations considered are Earth’s non-spherical gravity field, atmospheric drag, solar radiation pressure (SRP), third-body gravitational effects from the Sun and Moon, and relativistic corrections.¹⁸

The total force acting on the debris is given by:

$$\mathbf{F}_{total} = \mathbf{F}_{gravity} + \mathbf{F}_{drag} + \mathbf{F}_{SRP} + \mathbf{F}_{third-body} + \mathbf{F}_{relativity} \quad (2.2)$$

Numerical integration of the resulting equations of motion enables propagation of the state vector to predict future position and velocity.

2.2 Principles of Orbit Determination

Orbit Determination (OD) estimates an object’s state by fitting a dynamical model (i.e. the equations of motion including gravitational and perturbation forces) to observations. In this study, the initial state vector $\mathbf{x}_0 = [\mathbf{r}_0, \mathbf{v}_0]^T$ is obtained from a propagated TLE and iteratively refined using SDLR measurements. The resulting best-fit state $\mathbf{x}^*$ can then be propagated to assess prediction accuracy.

2.2.1 Measurement Modelling

For SDLR, the observable is the distance between the ground station and debris giving the observed range δ_{obs} :

$$\delta_{obs} = \frac{c \Delta t}{2} \quad (2.3)$$

SDLR measurements are typically distributed by the International Laser Ranging Service (ILRS) as Normal Points (NPT), representing temporally averaged range observations derived from full-rate data to reduce noise and data volume while preserving high precision.^{19,20}

At each iteration, the debris position $\mathbf{r}_{debris}$ is used to determine the computed range δ_{comp} across each observation epoch using the ground station position $\mathbf{r}_{station}$:

$$\delta_{comp} = \|\mathbf{r}_{debris} - \mathbf{r}_{station}\| \quad (2.4)$$

where both positions are expressed in a common reference frame (the station position is transformed from Earth-fixed ITRF to the inertial GCRF). The *observed–computed* residual at each epoch is then:

$$\Delta\delta = \delta_{obs} - \delta_{comp} \quad (2.5)$$

OD seeks to minimise these residuals across all observations by iteratively adjusting the state.

2.2.2 Batch Least Squares Estimation

The state is estimated using a Batch Least Squares (BLS) estimator, which minimises the weighted sum of squared residuals across all measurements.²¹ Batch Least Squares (BLS) is standard for LEO OD^{22,23} and has demonstrated centimetre-level accuracy for cooperative satellites.²⁴

For N measurements, the optimisation problem is:

$$J(\mathbf{x}) = \sum_{i=1}^N w_i (\delta_{obs,i} - \delta_{comp,i})^2 \quad (2.6)$$

where $w_i = 1/\sigma_i^2$ is each measurement weight. Here, equal weights are assumed.

The solution is obtained iteratively using the measurement Jacobian $\mathbf{H} = \partial\delta/\partial\mathbf{x}$, which relates state variations to measurement variations. Iteration continues until convergence is achieved.

This process also yields a covariance matrix $\mathbf{P}$, representing the uncertainty of the estimated state. This is transformed into the debris Local Vertical Local Horizontal (LVLH) frame to provide physically interpretable radial, along-track, and cross-track position uncertainties (defined in Figure 2.2).

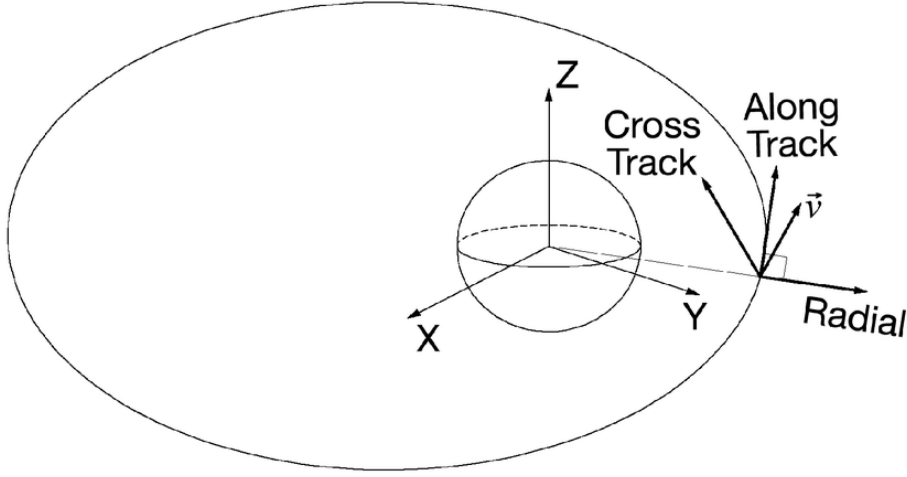


FIGURE 2.2. Definition of cross-track, along-track, and radial directions relative to an orbiting body in orbit about a central body. Taken from Benassai *et al.*, 2019.²⁵

2.2.3 Drag Parameter Estimation

Alongside the orbital state $\mathbf{x} = [\mathbf{r}, \mathbf{v}]^T$, the least-squares framework can be extended to estimate parameters within the force model. Here, the drag coefficient C_D is included in an augmented state vector:

$$\mathbf{x}_{aug} = [\mathbf{r}, \mathbf{v}, C_D]^T \quad (2.7)$$

This enables simultaneous estimation of orbit and drag. However, accurate estimation depends on parameter observability, i.e. the sensitivity of measurements to parameter variations. Using only SDLR, observability is limited: measurements are primarily sensitive to the radial direction (essentially providing one-dimensional characterisation along the laser track), while along-track position and velocity are weakly observed (Figure 2.3). As a result, C_D is hard for the estimator to separate from along-track position (Figure 2.2), as both produce similar signatures in the measurements.²¹

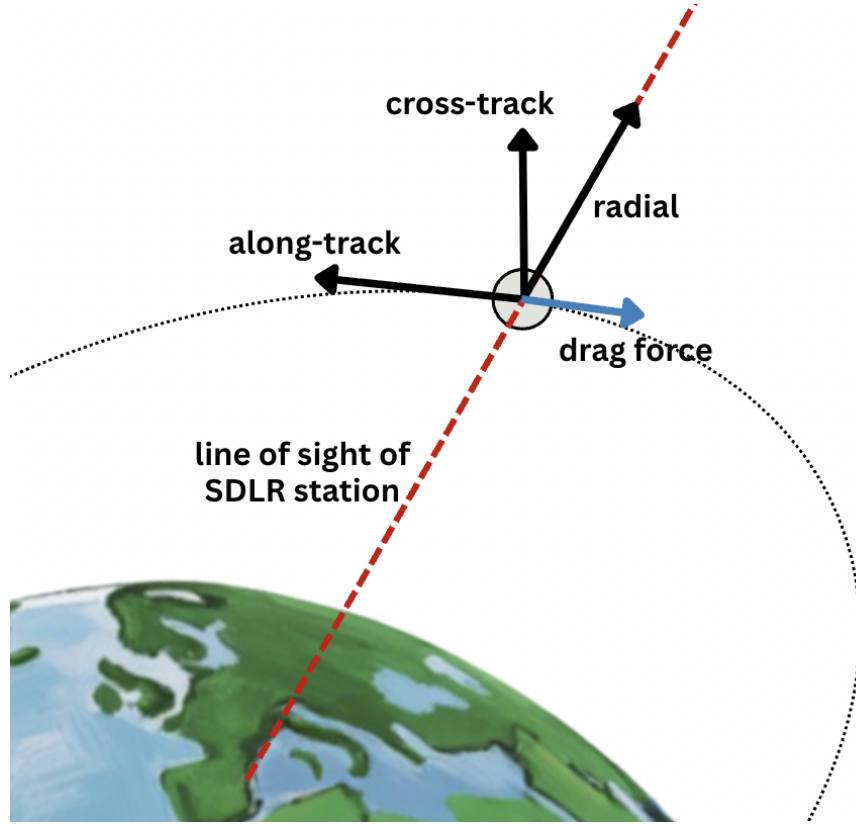


FIGURE 2.3. Illustration of parameter observability in SLR geometry. The radial direction aligns with the station line-of-sight (laser direction) and is therefore well observed. Along-track motion and drag act predominantly perpendicular to this line-of-sight, resulting in reduced observability and increased uncertainty.

2.3 Analysis Performance Metrics

OD accuracy is quantified using the root mean square (RMS) of residuals:

$$\text{RMS} = \sqrt{\frac{1}{N} \sum_{i=1}^N (\delta_{obs,i} - \delta_{comp,i})^2} \quad (2.8)$$

where $\delta_{obs,i}$ and $\delta_{comp,i}$ are the observed and computed range measurements at each of the N observations. RMS provides an intuitive measure of orbit fit quality in metres and is standard in orbit determination and laser ranging analysis.¹⁹

Prediction performance assessed through degradation/improvement factors, defined as the ratio of prediction RMS to OD and TLE RMS values. A factor of $1\times$ indicates no change in accuracy.

Position errors are analysed in the radial, along-track, and cross-track directions (Figure 2.2).

Chapter 3

Methodology

The methodology is shown in Figure 3.1. Three approaches are compared: (1) OD with fixed drag, (2) OD with C_D estimation, and (3) TLE propagation, evaluated over 2- and 7-day intervals to assess prediction quality.

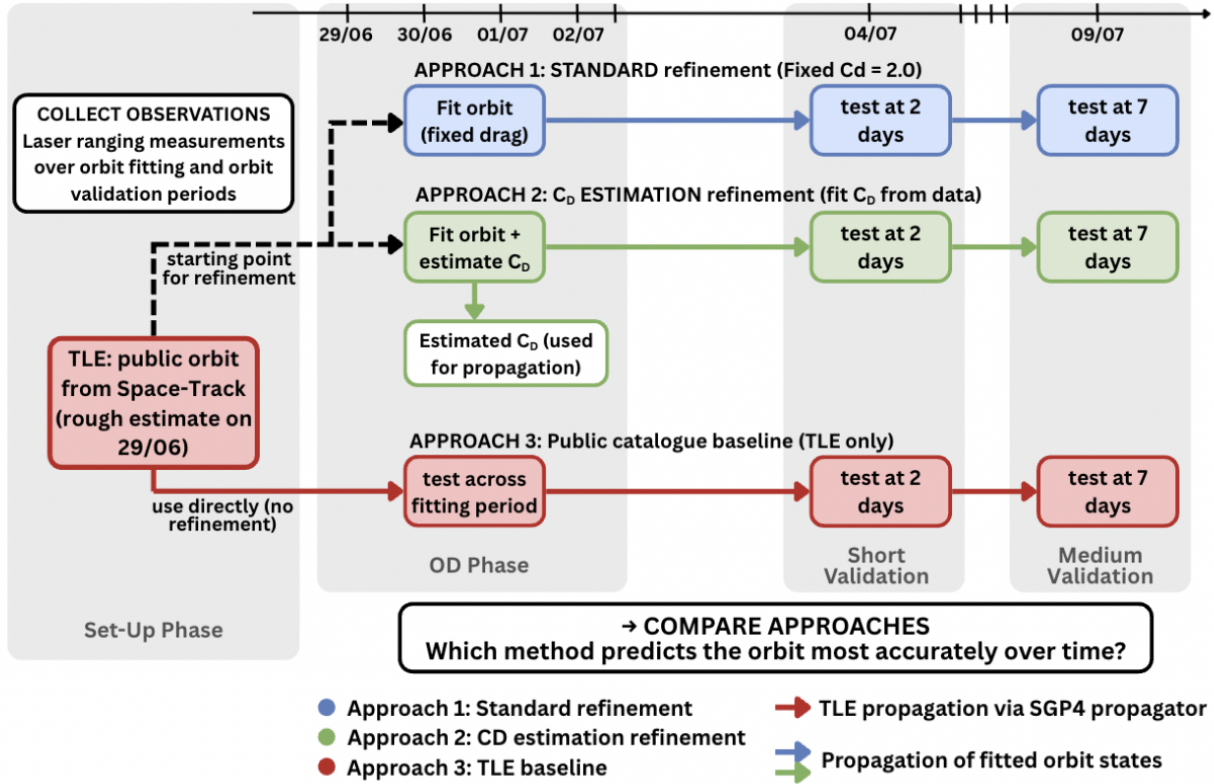


FIGURE 3.1. Orbit determination and validation methodology workflow. Laser ranging observations from 29 June–2 July 2025 are used to fit orbits under three approaches: (1) baseline with fixed $C_D = 2.0$ (blue), (2) C_D estimation where drag coefficient is fitted from data (green), and (3) TLE propagation without refinement (red). The SLR-based approaches initialise from a TLE orbit estimate (dashed arrows) which is refined using batch least squares estimation, while the TLE approach propagates the catalog orbit independently (solid red arrow). Each approach is validated against independent observations at 2 days (4 July) and 7 days (9 July) to assess prediction accuracies.

3.1 Set-Up Phase: Data Sources

3.1.1 SLR Measurements

SDLR observations were obtained for two debris objects from the Izaña-2 station²⁶ (Table 3.1). Measurements are provided as NPT files in CRD format, which contains information supporting the measurements such as the atmospheric conditions at the time of measurement.²⁰

For each object, datasets are split into training (OD phase) and prediction validation periods. A measurement uncertainty of $\sigma = 1.0$ m is assumed, consistent with reported SDLR precision.¹⁴ Tropospheric delay corrections, accounting for the signal passing through a greater thickness of atmosphere at low elevation angles, are included in the provided measurements. Observations below 10° elevation were excluded to limit modelling errors.

TABLE 3.1. Specifications of the Izaña-2 SDLR/SLR station

Parameter	Value
ILRS Identifier	7701
Station Name	Izaña-2
Location	Tenerife, Canary Islands, Spain
Latitude	28.29959791° N
Longitude	16.5106116° W
Altitude	2429.662 m
Reference Frame	ITRF2014
Operational Since	2003

3.1.2 Two-Line Elements (TLEs)

TLEs are publicly available and are sourced from Space-Track. They are used to initialise the state vector $\mathbf{x}_0$ and provide a baseline for comparison via standard SGP4 propagation.

3.1.3 Study Objects

Two space debris objects are used in this analysis, their characteristics²⁷ and orbital parameters are summarised in Table 3.2.

TABLE 3.2. Study object characteristics and orbital parameters

Parameter	Object 2123 (OV1-5 R/B)	Object 2680 (Diadème 1)
NORAD Catalog ID	2123	2680
International Designator	1965-082E	1967-011A
Object Type	Rocket body	Satellite
Launch Year	1966	1967
Mass (kg)	125	23
Cross-sectional Area (m ²)	1.347	0.455
Area-to-Mass Ratio (m ² /kg)	0.011	0.020
<i>Orbital Parameters (June 2025)</i>		
Semi-major Axis (km)	7399	7489
Eccentricity	0.005	0.069
Inclination (°)	144.6	39.4
Perigee Altitude (km)	1040	603
Apogee Altitude (km)	1060	1635
Altitude Variation (km)	20	1032
Orbital Period (min)	114.5	118.2
Orbit Classification	Near-circular LEO	Eccentric LEO

The data coverage of these two objects is described in Table 3.3. Both objects have large training datasets. Object 2123 has one short validation dataset while Object 2680 has 2 validation datasets with more measurements.

TABLE 3.3. Summary of SDLR observation datasets coverage

Dataset	Object	Time Period	Passes	Meas.
Training	2123 (OV1-5 R/B)	29 Jun – 2 Jul 2025	~ 8	177
	2680 (Diadème 1)	29 Jun – 2 Jul 2025	~ 10	214
Validation	2123 (OV1-5 R/B)	4 Jul 2025 (short-term)	1 (108 s)	9
	2680 (Diadème 1)	4 Jul 2025 (short-term)	multiple	53
	2680 (Diadème 1)	9 Jul 2025 (medium-term)	—	5

Both are inactive, uncontrolled debris tracked by ILRS stations, with physical parameters documented by ESA,²⁷ making them suitable candidates for SDLR-based OD analysis.

Object 2123 is near-circular ($e = 0.005$) spanning $\approx 1050 \pm 10$ km altitude, experiencing near-constant atmospheric density, while Object 2680 is more eccentric ($e = 0.069$) and spans 603–1635 km altitude, enabling investigation across eccentricity regimes (RQ2, RQ3). Figure 3.2 shows each object.



(A) OV1: Orbiting Vehicle 1



(B) Diadème 1 geodetic mission

FIGURE 3.2. Study objects

3.1.4 Auxiliary Data

Auxiliary datasets required for force modelling and frame transformations are summarised below. (Table 3.4).

TABLE 3.4. Auxiliary datasets and constants used for force modelling and reference frame transformations, all sourced directly through the modelling software.

Dataset	Specification/Source	Purpose
Earth orientation	IERS EOP 14 C04	Frame transformations for GCRF $\leftrightarrow$ ITRF
Space weather	NRLMSISE-00 ²⁸ atmospheric density model (NOAA)	Drag calculations
Gravity field	EGM96, degree & order 64	Variations in Earth’s gravity field
Planetary ephemerides	JPL DE430	Sun and Moon position vectors, 3rd body effects
Leap seconds	UTC–TAI offset	Time scale transformations and corrections

Physical constants: $\mu_{\oplus} = 398600.4418 \text{ km}^3 \text{ s}^{-2}$, $R_{\oplus} = 6378.137 \text{ km}$, $c = 299792.458 \text{ km s}^{-1}$, loaded automatically at initialisation.

3.2 Orbit Determination Phase

3.2.1 Processing Workflow

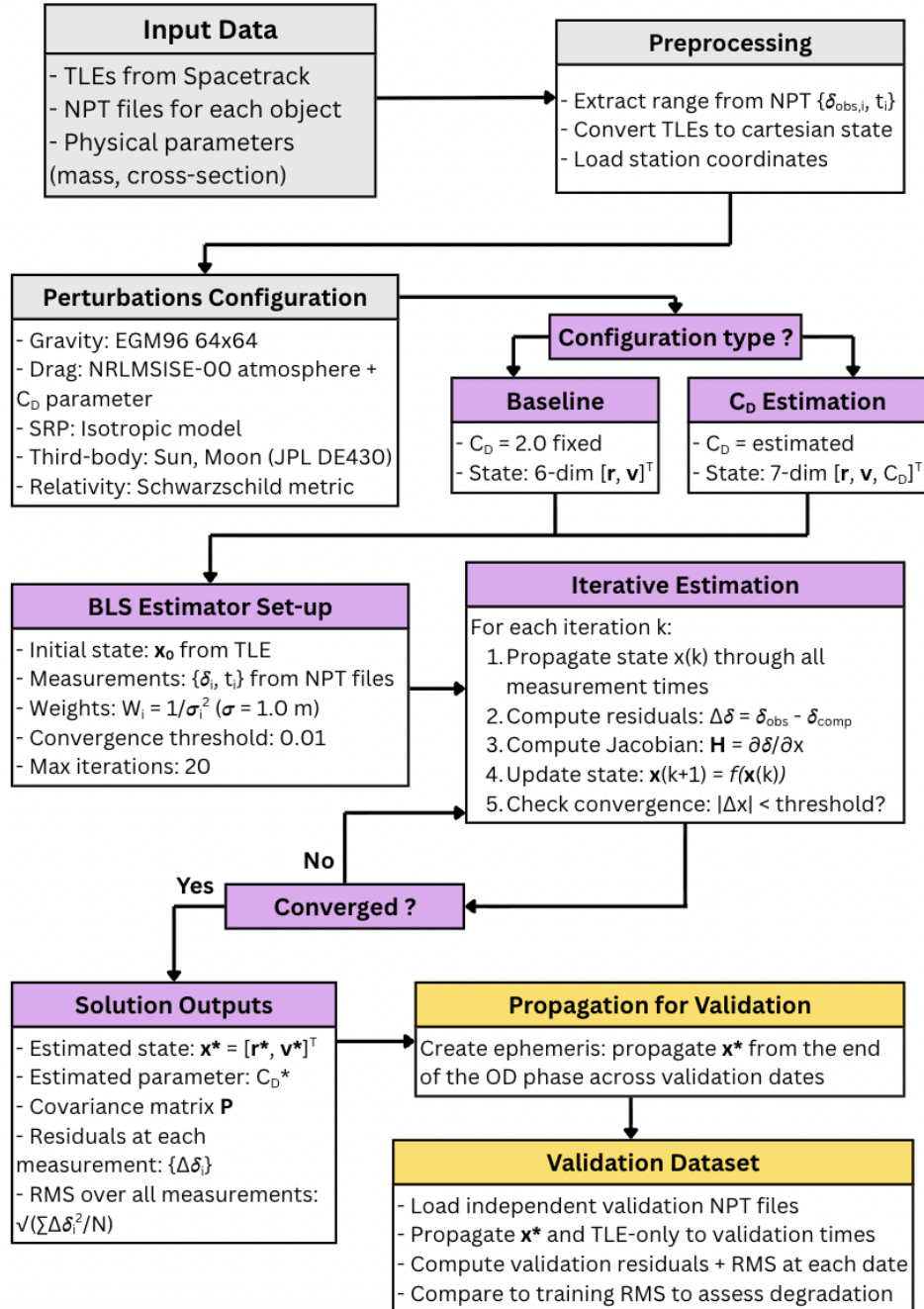


FIGURE 3.3. Orbit Determination Workflow Diagram. Grey = Set-up phase, pink = OD phase, yellow = validation phase

3.2.2 Software Implementation (Orekit 12.1)

Orbit determination is implemented in Python using Orekit (v12.1), an established astrodynamics library.²⁹ SDLR measurements are parsed to extract observed ranges, accounting for measurement timing conventions. The Izaña-2 station position is treated as fixed. Key numerical and modelling parameters are summarised in Table 3.5, with full implementation details provided in Appendix A.

TABLE 3.5. Software implementation parameters

Parameter	Description / Value
Numerical integrator	Variable step-size, max step 300 s
Estimator algorithm	Levenberg–Marquardt
Convergence threshold	10^{-2}
Maximum iterations	20
Measurement uncertainty	$\sigma = 1.0$ m

3.2.3 Estimation Configurations

Two configurations are compared to assess the impact of drag uncertainty:

TABLE 3.6. Estimation configuration parameters

Parameter	Baseline	C_D Estimation
State vector dimension	6	7
State vector components	$[\mathbf{r}, \mathbf{v}]^T$	$[\mathbf{r}, \mathbf{v}, C_D]^T$
C_D	Fixed at 2.0	Estimated (initial: 2.0)

For the C_D estimation configuration, the resulting position uncertainties in the radial, along-track, and cross-track directions are reported as a secondary output alongside RMS, as they are indicative of the correlation between C_D and along-track position discussed in Section 2.2.3.

3.3 Validation Phase

Validation is performed by propagating the fitted state beyond the training period using Orekit’s built-in propagators. Residuals are computed against independent SDLR measurements at 2-day (short-term) and 7-day (medium-term) intervals (Figure 3.4).

Prediction degradation is quantified by comparing validation RMS to training RMS. TLE-based predictions (SGP4) are evaluated against the data over the same intervals to provide a baseline (RQ1, RQ3).

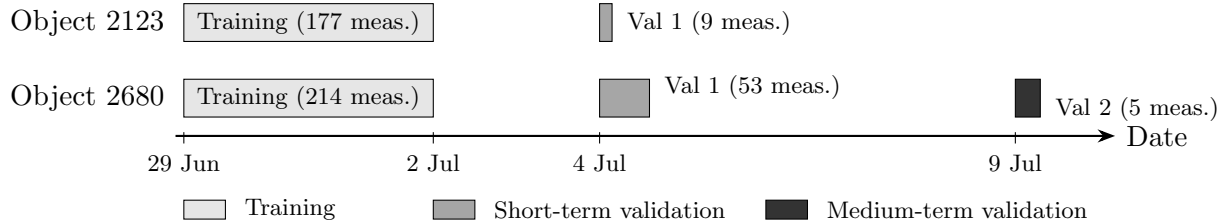


FIGURE 3.4. Temporal structure of training and validation datasets for both study objects. Block widths are proportional to the time span.

Chapter 4

Results

4.1 Object 2123 (OV1-5 R/B, $e = 0.005$)

4.1.1 Orbit Determination Phase

Baseline Training Performance

The baseline configuration achieved a training RMS of 3.3 m over 177 measurements. Results are summarised in Table 4.1 and Figures 4.1 and 4.2.

TABLE 4.1. Object 2123 baseline training results

Parameter	Value
RMS	3.3 m
Number of measurements	177
Convergence iterations	12
Mean residual	2.1 m
Residual std. dev.	3.3 m

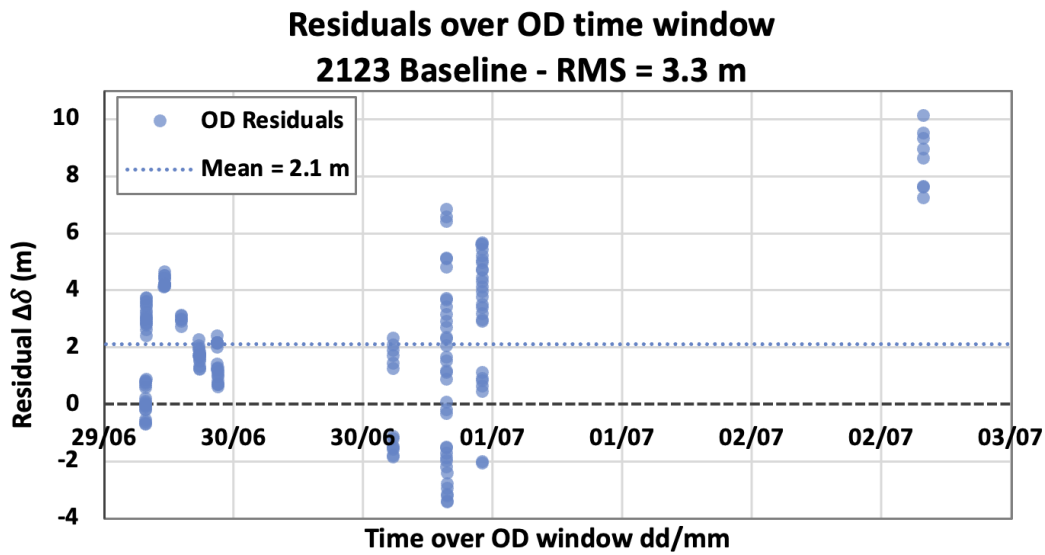


FIGURE 4.1. Object 2123 baseline OD residuals versus time over the training arc (29 June – 2 July 2025), showing pass dates. RMS = 3.3 m. Residuals show no systematic temporal trend, with pass-to-pass scatter reflecting varying observation geometry.

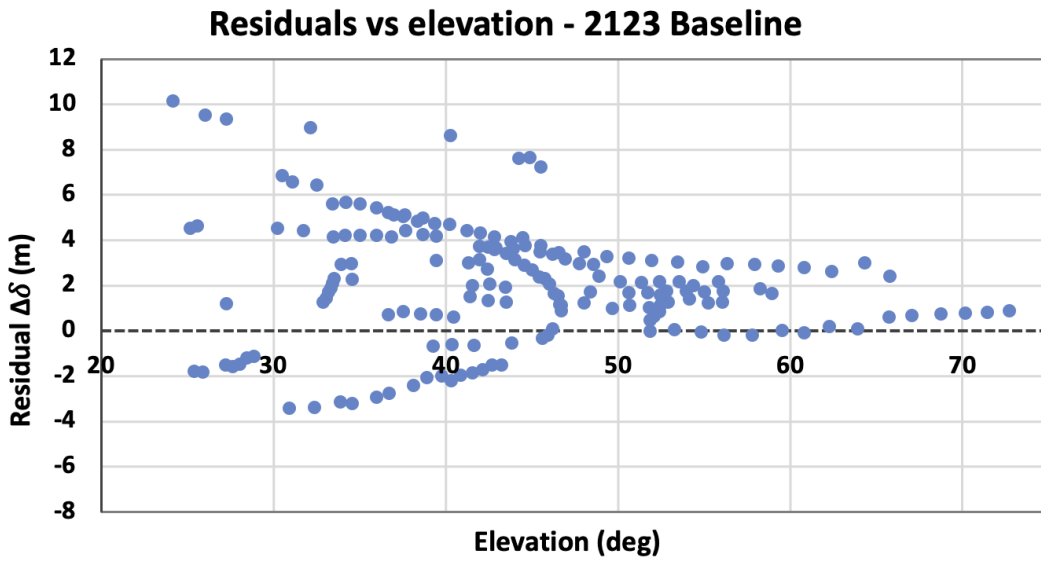


FIGURE 4.2. Object 2123 baseline OD residuals versus elevation angle. There are larger residuals at low elevation angles ($< 30^\circ$).

C_D Estimation Training Performance

Enabling C_D estimation reduced training RMS from 3.3 m to 1.2 m, a 63% improvement. Results are summarised in Table 4.2 and Figures 4.3 and 4.4.

TABLE 4.2. Object 2123 C_D estimation training results

Parameter	Value
Training RMS	1.2 m
Number of measurements	177
Convergence iterations	5
Mean residual	0.7 m
Residual std. dev.	1.0 m
C_D (estimated)	1.628 ± 0.015
Change from initial	-18.6%

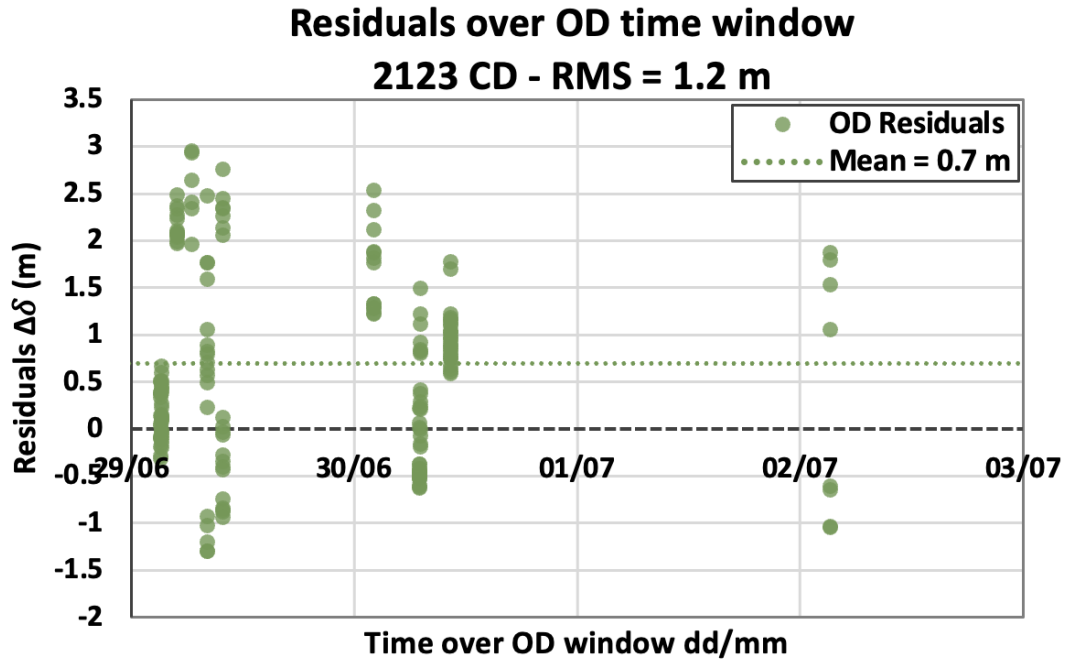


FIGURE 4.3. Object 2123 C_D estimation OD residuals versus time. RMS = 1.2 m. The reduced scatter compared to Figure 4.1 reflects the improved force model fit achieved by estimating C_D .

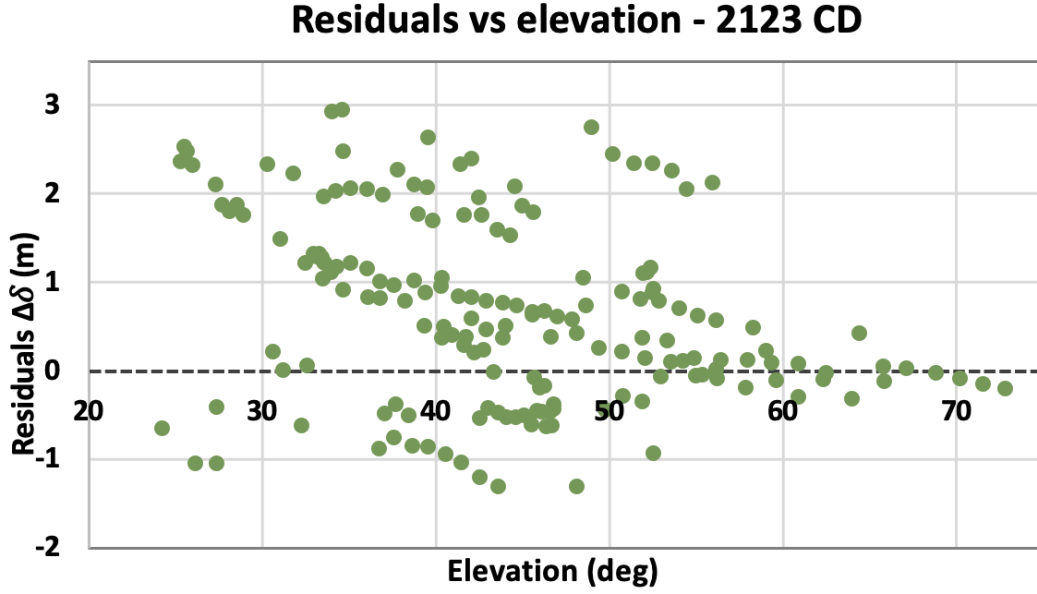


FIGURE 4.4. Object 2123 C_D estimation OD residuals versus elevation angle. Elevation dependency is substantially reduced compared to the baseline (Figure 4.2), with low-elevation residuals contained within ± 3 m.

Position Uncertainty Analysis

Table 4.3 compares position uncertainties between configurations. Enabling C_D estimation shows radial improvement ($0.844 \text{ m} \rightarrow 0.221 \text{ m}$) and along-track degradation ($0.260 \text{ m} \rightarrow 2.791 \text{ m}$).

TABLE 4.3. Object 2123 position uncertainties by configuration (1σ)

Component	Baseline	C_D Estimation
	Pos. (m)	Pos. (m)
Radial	0.844	0.221
Along-track	0.260	2.791
Cross-track	0.383	0.404

4.1.2 Validation Phase

Short-term Validation (4 July 2025)

Validation was performed on a single pass of 108 s comprising 9 measurements. Results are shown in Table 4.4.

TABLE 4.4. Object 2123 short-term validation results (4 July 2025)

Metric	Baseline	C_D Estimation
Validation RMS	0.7 m	22.2 m
TLE RMS		432.2 m
Improvement over TLE	617×	19.5×

Figures 4.5 and 4.6 show the validation residuals for both configurations alongside TLE prediction errors. TLE errors decrease across the pass, while SLR-OD errors remain near zero for the baseline and within 30 m for the C_D estimation.

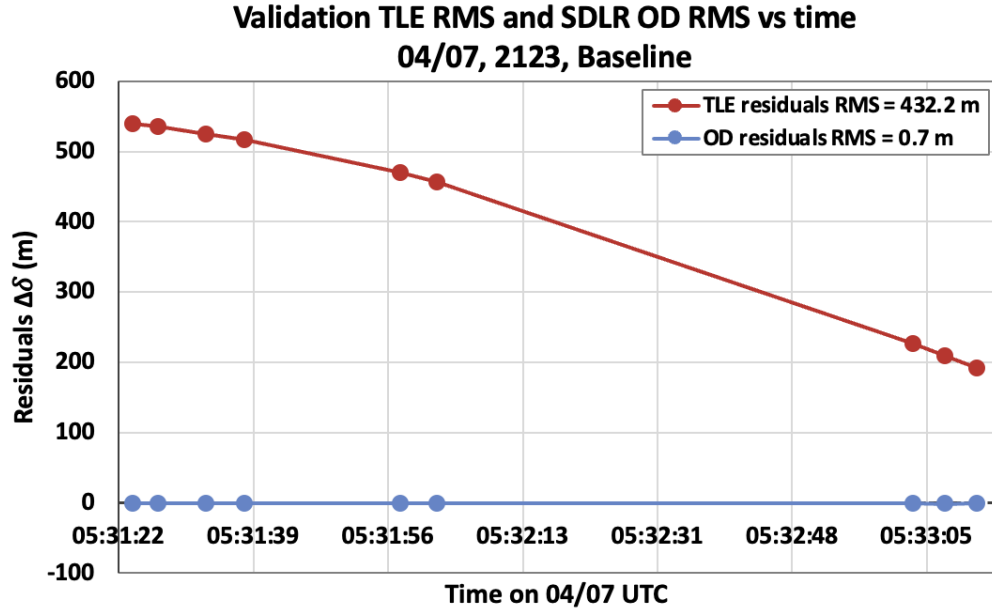


FIGURE 4.5. Object 2123 baseline configuration: SLR-OD and TLE prediction residuals versus time on 4 July 2025. SLR-OD residuals (RMS = 0.7 m) remain near zero throughout the pass while TLE errors decrease from ~ 535 m to ~ 190 m across the pass.

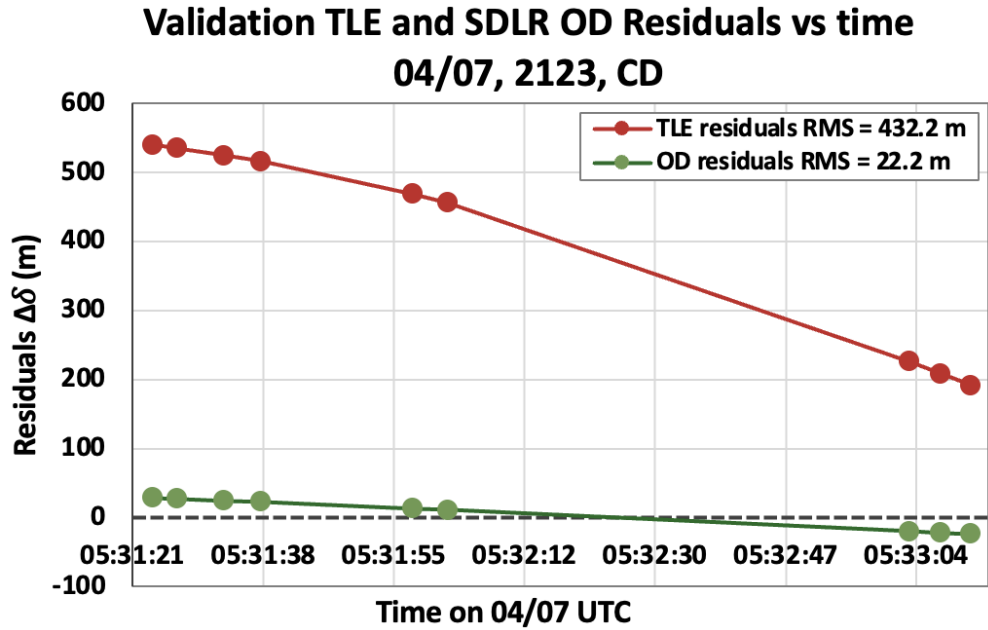


FIGURE 4.6. Object 2123 C_D estimation configuration: SLR-OD and TLE prediction residuals versus time on 4 July 2025. SLR-OD residuals (RMS = 22.2 m) show a monotonic drift from ~ 30 m to ~ -10 m across the pass, indicative of a growing along-track error. TLE errors follow the same pattern as Figure 4.5.

Figure 4.7 summarises training and validation RMS for both configurations alongside TLE, showing the improvement achieved by SLR-OD over the training arc and the divergence between configurations at validation.

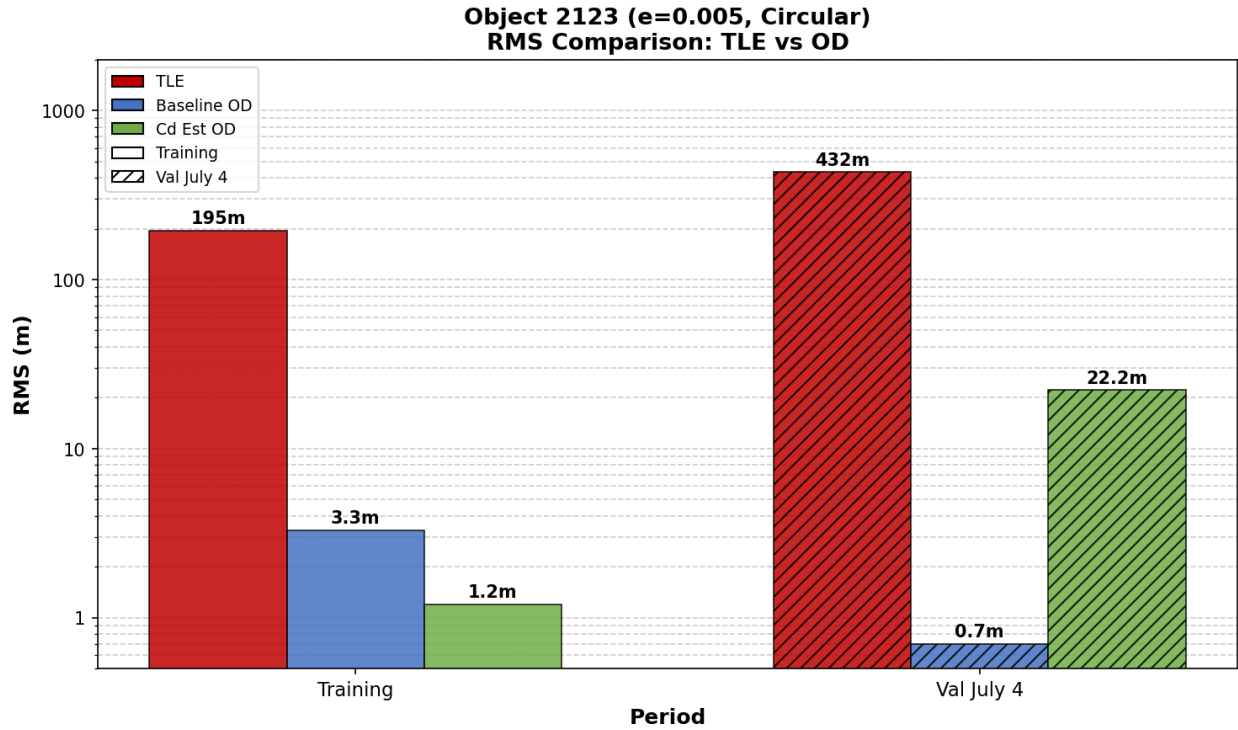


FIGURE 4.7. Object 2123: RMS comparison between TLE, baseline OD, and C_D estimation OD for the training phase and short-term validation (4 July 2025). Log scale used to accommodate the dynamic range. The baseline validation result (0.7 m) is anomalous.

4.2 Object 2680 (Diadème 1, $e = 0.069$)

4.2.1 Orbit Determination Phase

Baseline Training Performance

The baseline configuration achieved a training RMS of 27.1 m over 214 measurements. Results are summarised in Table 4.5 and Figures 4.8 and 4.9.

TABLE 4.5. Object 2680 baseline training results

Parameter	Value
RMS	27.1 m
Number of measurements	214
Convergence iterations	15
Mean residual	-1.9 m
Residual std. dev.	27.0 m

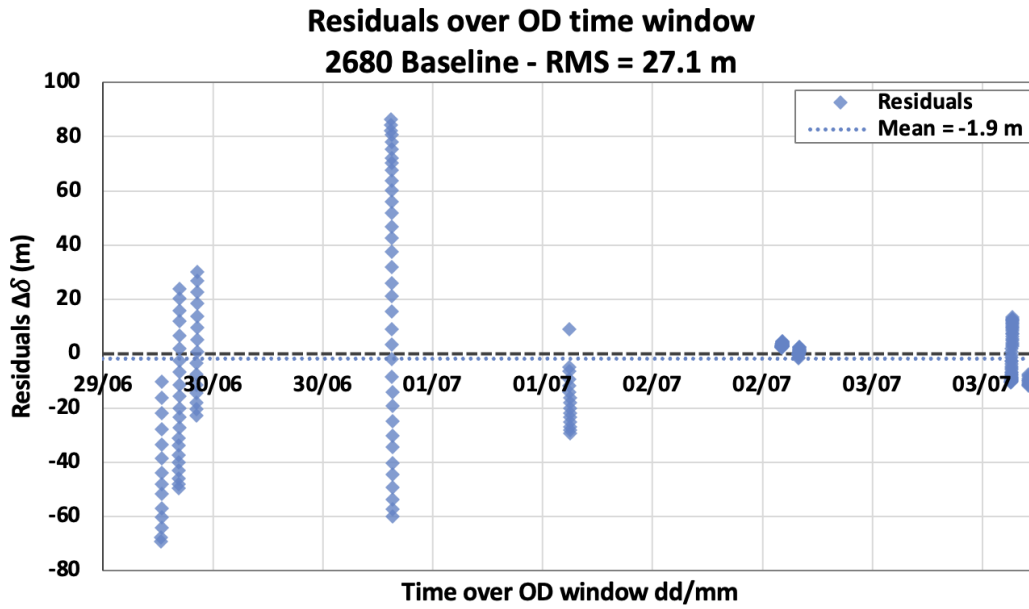


FIGURE 4.8. Object 2680 baseline OD residuals versus time over the training arc (29 June – 2 July 2025). RMS = 27.1 m. Large pass-to-pass variation is visible.

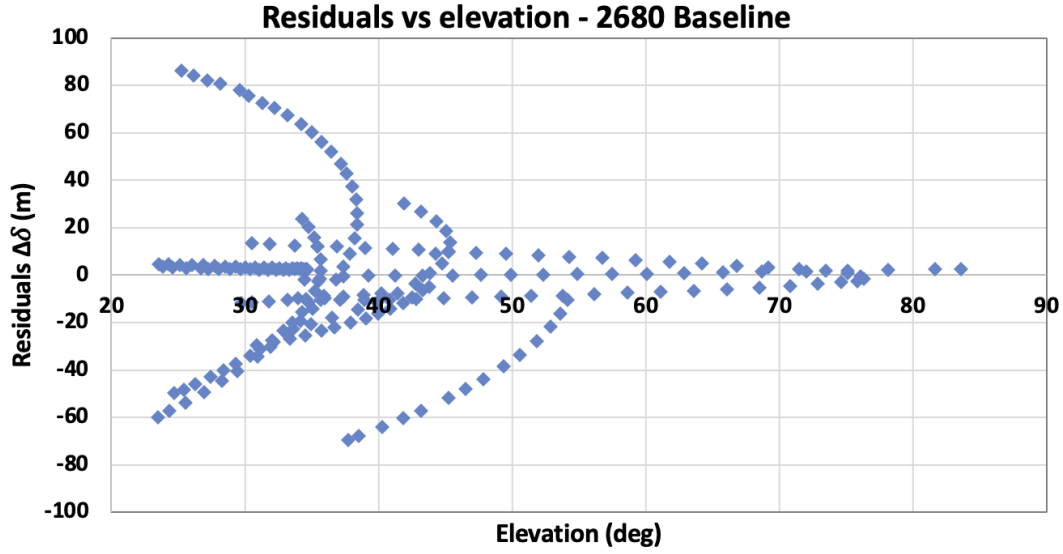


FIGURE 4.9. Object 2680 baseline OD residuals versus elevation angle. A strong elevation dependency is visible, with residuals reaching ± 80 m at low elevations.

C_D Estimation Training Performance

Enabling C_D estimation reduced training RMS from 27.1 m to 17.7 m, a 35% improvement. Results are summarised in Table 4.6 and Figures 4.10 and 4.11.

TABLE 4.6. Object 2680 C_D estimation training results

Parameter	Value
Training RMS	17.7 m
Number of measurements	214
Convergence iterations	5
Mean residual	-2.1 m
Residual std. dev.	17.6 m
C_D (estimated)	1.228 ± 0.001
Change from initial	-38.6%

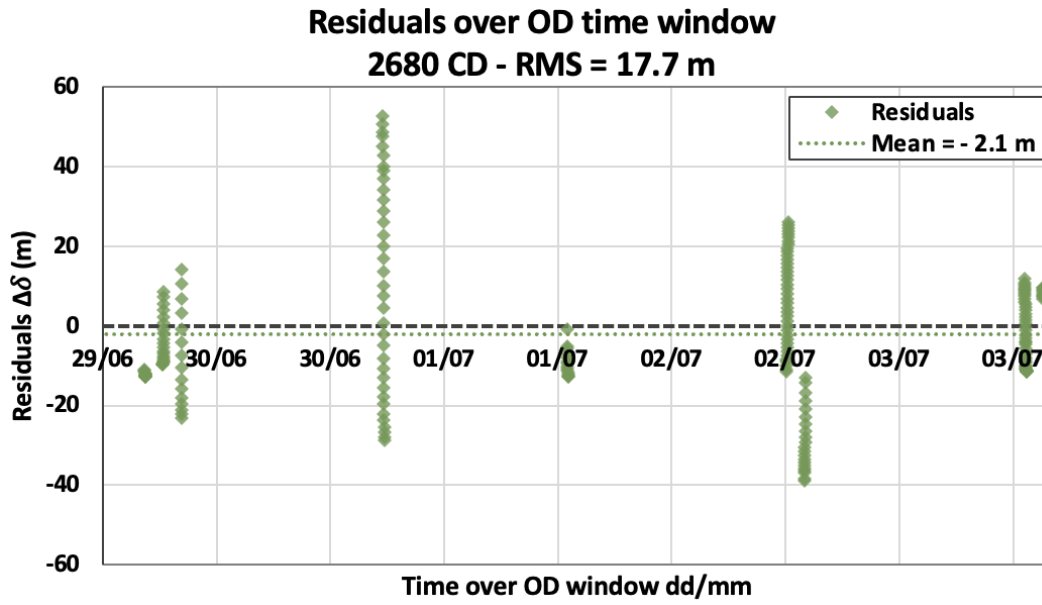


FIGURE 4.10. Object 2680 C_D estimation OD residuals versus time. RMS = 17.7 m. Pass-to-pass variation is reduced compared to Figure 4.8. Residuals remain large for some passes.

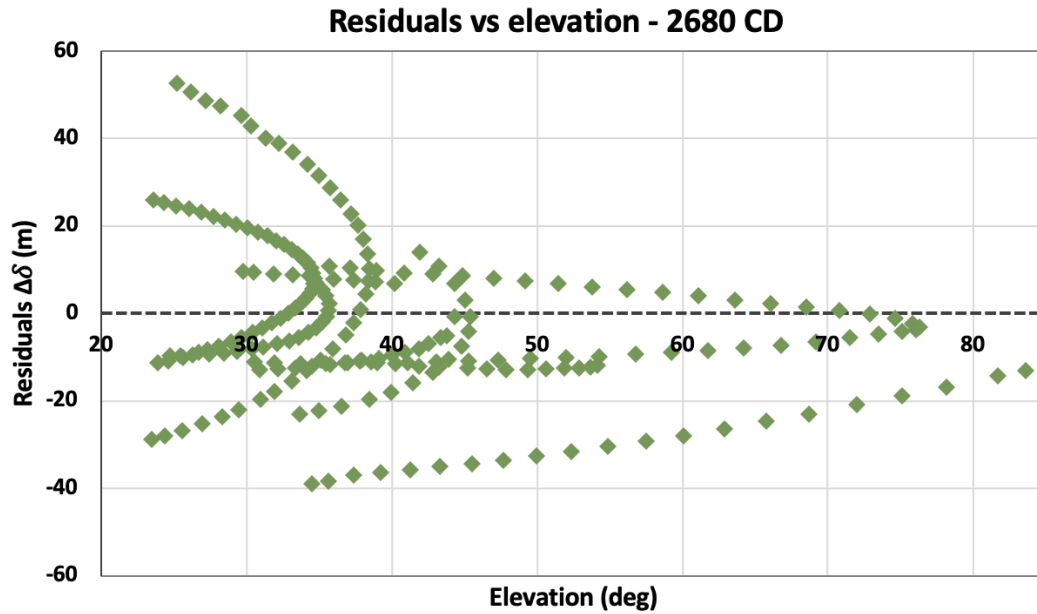


FIGURE 4.11. Object 2680 C_D estimation OD residuals versus elevation angle. The fan-shaped elevation dependency is partially reduced compared to the baseline (Figure 4.9), though significant error remains at low elevations.

Position Uncertainty Analysis

Table 4.7 compares position uncertainties between configurations. The same radial improvement and along-track degradation pattern is observed, but with reduced magnitude compared to Object 2123.

TABLE 4.7. Object 2680 position uncertainties by configuration (1σ)

	Baseline (m)	C_D Estimation (m)
Radial	0.437	0.226
Along-track	0.565	1.138
Cross-track	0.322	0.324

4.2.2 Validation Phase

Short-term Validation (4 July 2025)

Short-term validation was performed across 53 measurements over multiple passes. Results are shown in Table 4.8.

TABLE 4.8. Object 2680 short-term validation results (4 July 2025)

Metric	Baseline	C_D Estimation
Validation RMS	329.8 m	239.2 m
Degradation factor	12.2×	13.5×
TLE RMS		1291.4 m
Improvement over TLE	3.9×	5.4×

Both configurations degrade relative to training RMS but remain below TLE error. Figures 4.12 and 4.13 show the validation residuals versus time.

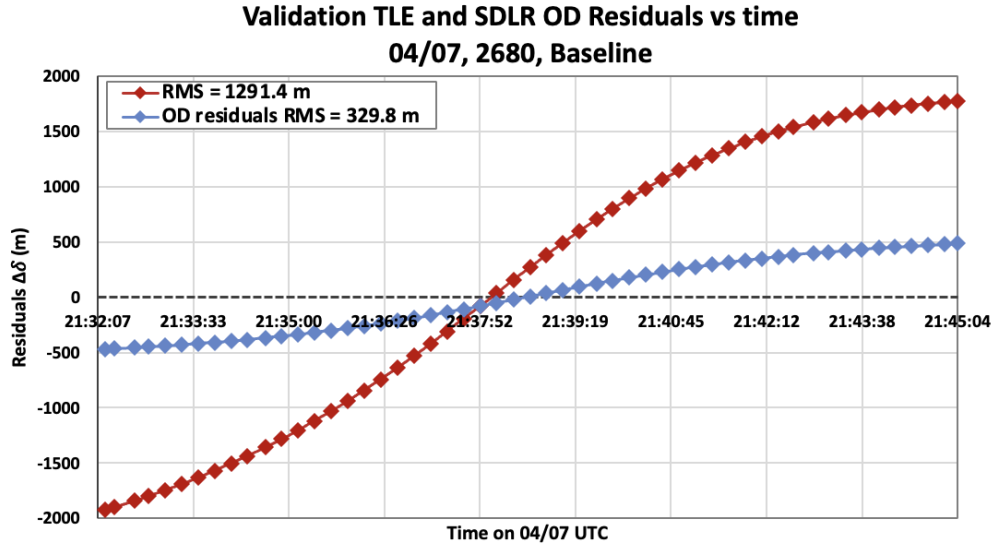


FIGURE 4.12. Object 2680 baseline configuration: SLR-OD and TLE prediction residuals versus time on 4 July 2025. Both show a drift pattern across the pass. SLR-OD (RMS = 329.8 m) substantially outperforms TLE (RMS = 1291.4 m).

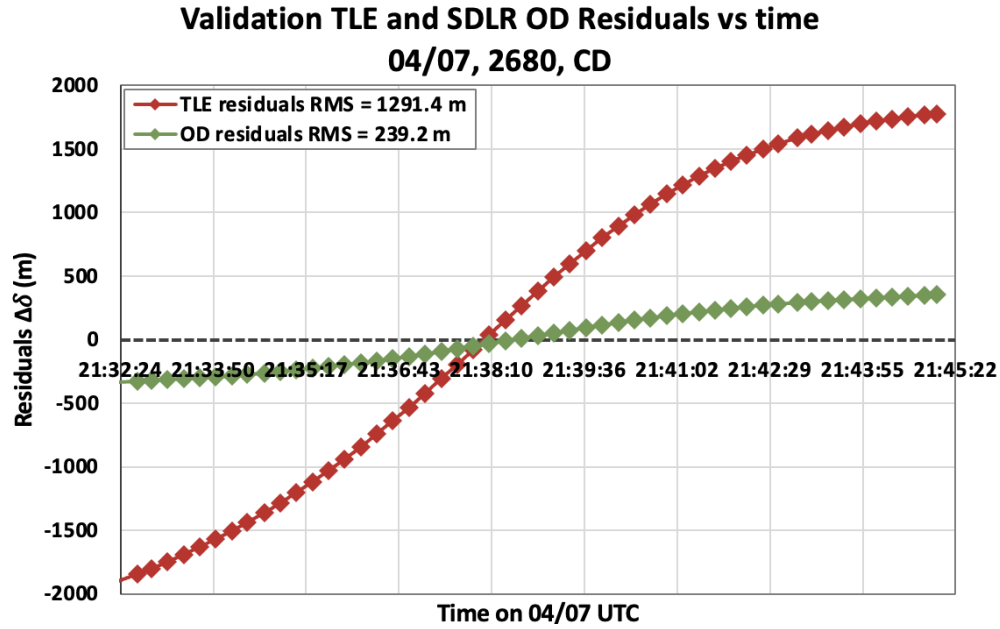


FIGURE 4.13. Object 2680 C_D estimation configuration: SLR-OD and TLE prediction residuals versus time on 4 July 2025. SLR-OD (RMS = 239.2 m) shows a similar drift pattern to the baseline but with reduced magnitude. Both configurations are substantially improved over TLE.

Medium-term Validation (9 July 2025)

Medium-term validation comprises 5 measurements. Results are shown in Table 4.9.

TABLE 4.9. Object 2680 medium-term validation results (9 July 2025)

Metric	Baseline	C_D Estimation
Validation RMS	140.4 m	1221.9 m
Degradation factor	5.2×	69.0×
TLE RMS	3208.2 m	
Improvement over TLE	22.9×	2.6×

Performance diverges significantly between configurations at 7 days. The C_D estimation configuration produces a large offset. Figures 4.14 and 4.15 show the residuals. Baseline residuals remain approximately +100 m, while C_D estimation residuals are approximately +1200 m. TLE errors span -4000 m to -2600 m, within expected range.

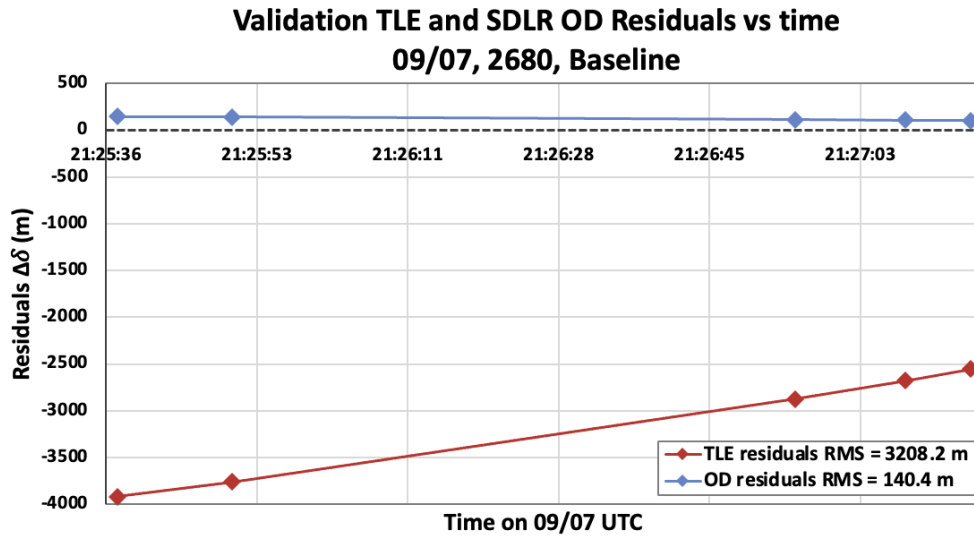


FIGURE 4.14. Object 2680 baseline configuration: SLR-OD and TLE prediction residuals versus time on 9 July 2025.

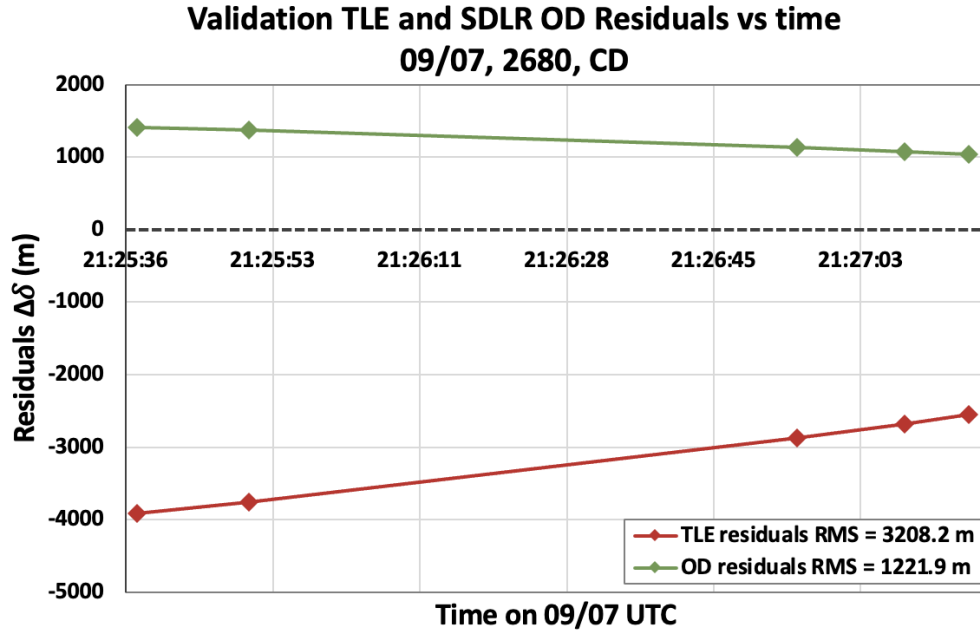


FIGURE 4.15. Object 2680 C_D estimation configuration: SLR-OD and TLE prediction residuals versus time on 9 July 2025.

Figure 4.16 summarises training and validation RMS across all phases and configurations, showing the divergence between baseline and C_D estimation at medium-term validation and the consistent improvement over TLE.

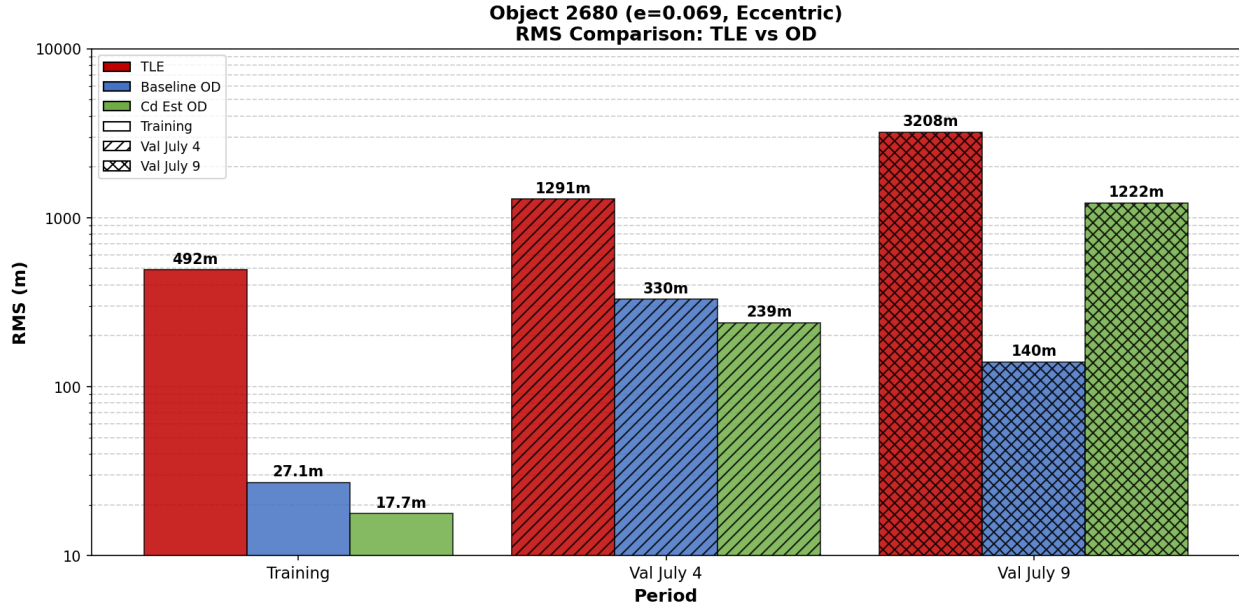


FIGURE 4.16. Object 2680: RMS comparison between TLE, baseline OD, and C_D estimation OD across the training phase and both validation dates. Log scale used to accommodate the dynamic range. At 7 days, C_D estimation (1222 m) degrades catastrophically while baseline (140 m) maintains a $22.9\times$ improvement over TLE (3208 m).

4.3 Summary of Key Findings

4.3.1 Numerical Results

Table 4.10 summarises all numerical results.

TABLE 4.10. Complete RMS results summary: training and validation, all configurations

Object	Phase	TLE RMS	Baseline RMS	C_D Est. RMS	Est. C_D
2123	OD	195 m	3.3 m	1.2 m	1.628
	4 Jul val.*	432.2 m	0.7 m*	22.2 m	—
2680	OD	492 m	27.1 m	17.7 m	1.228
	4 Jul val.	1291.4 m	329.8 m	239.2 m	—
	9 Jul val.	3208.2 m	140.4 m	1221.9 m*	—
* Anomalous					

4.3.2 Principal Findings

The following findings address the three research questions:

- (1) **RQ1 – SDLR vs TLE:** SDLR-based OD achieved training improvement factors of $18\times$ – $163\times$ and validation improvement factors of $2.6\times$ – $617\times$ over TLE propagation, confirming operationally meaningful orbit knowledge beyond the public catalogue across all configurations and timescales.
- (2) **RQ2 – Effect of C_D estimation:** C_D estimation improved training fit by 63% (Object 2123, 1.628 ± 0.015) and 35% (Object 2680, 1.228 ± 0.001).
- (3) **RQ3 – Accuracy degradation:** Prediction accuracy degrades significantly beyond the training arc. Most critically, C_D estimation for Object 2680 exhibited catastrophic degradation at 7 days (1221.9 m, $69\times$ degradation), an anomalous result. The Object 2123 baseline validation result (0.7 m) is also anomalous.

Chapter 5

Discussion

5.1 Orbit Determination Phase Performance (RQ1)

Baseline SDLR-based OD without C_D estimation achieved training RMS of 3.3 m (Object 2123) and 27.1 m (Object 2680), representing $59\times$ and $18\times$ improvements over TLE propagation respectively. The metre-level accuracy achieved for Object 2123 is consistent with published SDLR-based OD results for LEO objects,¹⁴ confirming that single-station SDLR can meaningfully improve on TLE-derived orbit knowledge. Both objects show that SDLR-based OD outperforms TLE across all phases, with improvements of $18\text{--}163\times$ during training and $2.6\text{--}617\times$ during validation (Figure 4.7, 4.16).

The difference in baseline RMS reflects contrasting atmospheric environments. Object 2123 experiences near-constant density (1050 ± 10 km altitude), while Object 2680’s eccentric orbit spans 603–1635 km and introduces large density variations (exceeding two orders of magnitude within a single orbital period¹⁸). The baseline $C_D = 2.0$ is a reasonable approximation for the near-circular case but introduces systematic residuals for the eccentric case as the model cannot represent the variation in atmospheric conditions. This contributes to the characteristic fan-shaped elevation dependency visible in Figure 4.9 alongside significant pass-to-pass variation in the 2680 residuals (Figures 4.8 & 4.10).

For ARMA operations, single-station SDLR OD can replace TLE-based targeting, but performance depends on orbital regime, requiring different strategies for circular and eccentric targets.

5.2 Drag Coefficient Estimation (RQ2)

5.2.1 Estimated Values and Observability

C_D estimation converged to 1.628 ± 0.015 for 2123 and 1.228 ± 0.001 for 2680 (both < 2.0). The estimated C_D values fall within physically expected ranges for LEO debris (1.0–3.0).²¹

The tighter uncertainty for Object 2680 reflects stronger drag observability from its larger atmospheric density variation, producing measurable changes in the orbit that the estimator can attribute to drag. 2123’s near-constant altitude provides little drag variation, reducing separability from other errors.

5.2.2 The C_D /Along-Track Covariance Trade-off

Along-track position and drag coefficient are strongly correlated due to their shared direction, producing similar, weakly observable error signatures which limit their separability in estimation.

This is visible in the position errors. For 2123, C_D estimation improves radial uncertainty from 0.844 m to 0.221 m while degrading along-track uncertainty from 0.260 m to 2.791 m, and for 2680, radial improves from 0.437 m to 0.226 m while along-track degrades from 0.565 m to 1.138 m. The smaller degradation for Object 2680 is consistent with its stronger drag observability.

In both cases, the estimator accepts degraded along-track precision in exchange for the improved high-observability radial precision and a better-defined drag model. This is expected

but important operationally, as along-track error dominates prediction degradation. Improving the observation geometry through multi-static ranging (multiple stations) would mitigate this trade-off.

5.2.3 Residual Patterns and Force Model Interpretation

Residual scatter over time is reduced compared to the baseline for both objects, reducing training RMS from 3.3 m to 1.2 m for Object 2123 and from 27.1 m to 17.7 m for object 2680. For Object 2123, C_D estimation substantially reduces the elevation dependency of residuals: the fan-shaped structure visible in Figure 4.2 collapses to a more uniform scatter in Figure 4.4, with low elevation errors reduced from ± 10 m to ± 3 m. Hence, much of the baseline elevation-dependent residual is attributable to incorrect C_D . A small positive bias of +0.7 m persists, likely reflecting residual tropospheric delay not fully corrected by the NPT preprocessing.

For Object 2680, C_D estimation only partially reduces the elevation dependency: the fan-shaped structure (Figure 4.9) persists in Figure 4.11, with low-elevation residuals reduced from ± 80 m to ± 55 m, confirming that a constant C_D cannot accurately represent an eccentric drag environment. The estimator converges to a weighted average that minimises total residuals but cannot eliminate the altitude-dependent signature.

The C_D estimation configurations converged faster (5 iterations for both objects) than baseline (12 and 15 iterations), despite the estimator handling more parameters. Since C_D estimation provides an additional degree of freedom, systematic errors are absorbed early in the iteration process, that would otherwise require multiple iterations of state vector adjustment to compensate.

5.2.4 Eccentricity as a Threshold Parameter

The contrasting behaviour of C_D estimation across the two objects suggest eccentricity may act as a threshold parameter for C_D estimation viability. Low eccentricity ($e < 0.01$) can

tolerate C_D estimation, while higher eccentricity ($e > 0.02$) introduces poor propagation (see Section 5.3). However, this is based on limited data and requires further validation.

Target selection for ARMA missions should consider orbital eccentricity as a key parameter when determining whether drag estimation techniques can be reliably applied.

5.3 Temporal Stability and Validation (RQ3)

Across all validation windows, TLE prediction errors show a characteristic monotonic drift within each validation pass (Figures 4.5 and 4.12), consistent with along-track error accumulation from SGP4’s simplified force models.

5.3.1 Object 2123 Short-term Validation

Object 2123’s validation result must be interpreted with caution. The single 108s pass comprising 9 measurements at 69° mean elevation is statistically insufficient for robust orbit accuracy assessment. The baseline result of 0.7 m, lower than the training RMS of 3.3 m, is physically implausible as a genuine accuracy improvement and is instead attributed to the highly favourable geometry of this specific pass. This pass provides a nearly radial viewing geometry in which range measurements are most sensitive. Under these conditions the RMS is artificially suppressed. The C_D estimation result of 22.2 m is more plausible but equally cannot be considered representative given the sample size. No meaningful conclusions about circular orbit prediction degradation can be drawn from this dataset.

5.3.2 Object 2680 Short-term Validation (2 Days)

Both configurations show comparable degradation at 2 days: 12.2× for baseline (329.8 m) and 13.5× for C_D estimation (239.2 m). The validation residual plots (Figures 4.12 and 4.13) show a characteristic drift pattern across the pass, transitioning from large negative to large positive residuals. This reflects along-track error accumulation: the propagated orbit lags the

true orbit, producing a range error that changes sign across the pass (negative on approach, positive on recession, Figure 5.1). Here, both configurations experience similar degradation as they are both limited by atmospheric uncertainty. The C_D estimation configuration maintains a modest advantage (RMS 239 m vs 330 m), suggesting the fitted C_D retains some validity 2 days after the orbit fit.

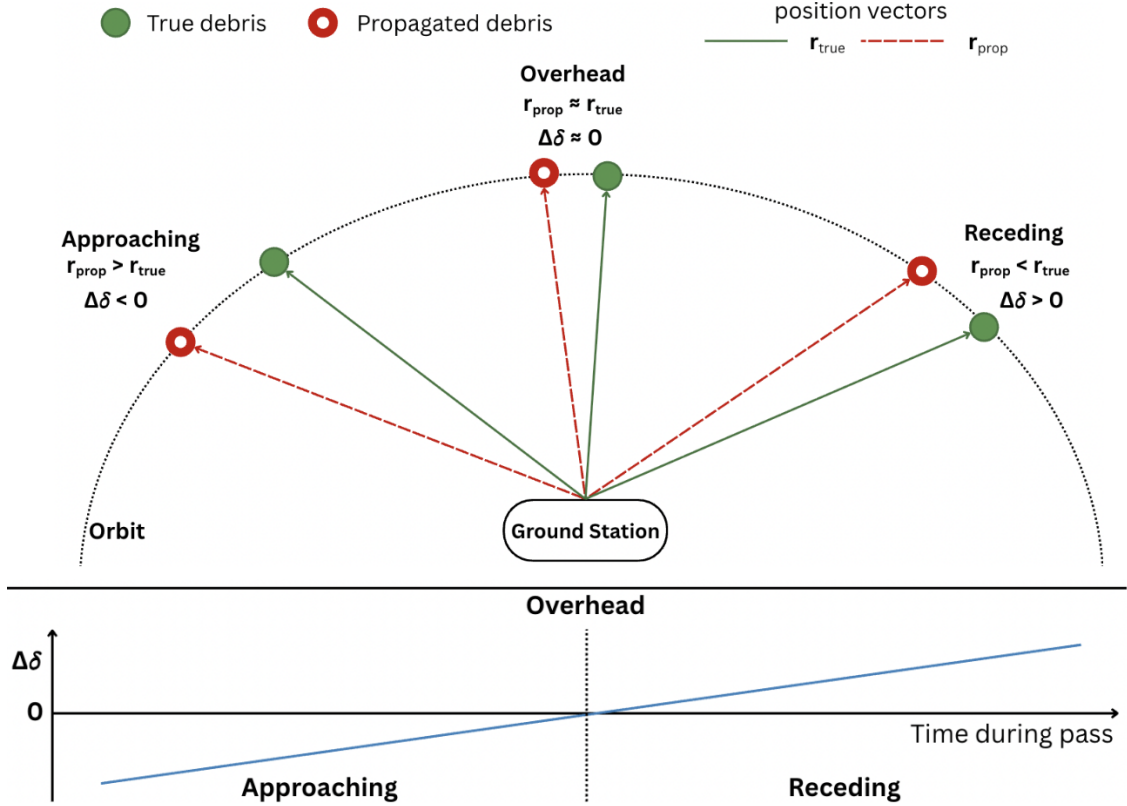


FIGURE 5.1. Origin of the residual drift observed in validation passes. The predicted position (open red circle) lags the true position (filled green circle) by an along-track offset due to accumulated force model error. Across a pass, this offset produces an observed range δ_{obs} (capturing true range) that transitions from shorter than the propagated range δ_{comp} (negative residual) to approximately equal (near-zero residual) to longer (positive residual), as the station line-of-sight rotates. The resulting $\Delta\delta = \delta_{obs} - \delta_{comp}$ drifts across the pass, producing the pattern visible in the residual plots on 04/07. This pattern is always expected with sufficient data.

5.3.3 Object 2680 Medium-term Validation (7 Days)

The divergence between configurations at 7 days is the most significant finding of this analysis. The baseline configuration achieves 140.4 m RMS ($5.2\times$ degradation), while C_D estimation produces 1221.9 m RMS with mean residual +1212.2 m nearly equal to the RMS, confirming that errors are dominated by a large systematic offset rather than random scatter. The propagated orbit has drifted to a fundamentally incorrect state.

This results from atmospheric evolution. The estimated $C_D = 1.228$ represents a weighted average optimised for the specific conditions observed over 29 June–2 July 2025. By 9 July, atmospheric conditions have evolved sufficiently that $C_D = 1.228$ no longer accurately represents the drag environment, making a period-specific constant C_D inappropriate for medium-term propagation.

The baseline configuration is more robust. The generic $C_D = 2.0$, though less accurate during any specific training window, represents a long-term average as opposed to transient atmospheric conditions. Hence, while it produces a higher training RMS, it degrades more slowly during propagation.

The low July 9 baseline result (140.4 m, lower than the July 4 result of 329.8 m) is likely in part a geometry effect. With only 5 measurements in a single short pass at favourable geometry, similar to 2123’s validation dataset, the RMS is not representative of the true orbit error.

For eccentric debris, fixed- C_D models provide more robust medium-term predictions, while fitted C_D values should not be used beyond short propagation windows.

5.4 Operational Implications for ARMA (RQ3)

5.4.1 Tracking Requirements by Orbit Type

The results indicate an eccentricity-dependent tracking approach.

For near-circular targets ($e < 0.01$), SDLR-based OD with C_D estimation achieves metre-level accuracy, satisfying the sub-100 m requirement for approach operations. Although validation for Object 2123 is limited, the stable atmospheric conditions at approximately 1050 km altitude suggest that multi-day tracking intervals may be feasible. However, this remains unverified. Within this regime, C_D estimation is beneficial due to its consistent improvement in orbit fit.

For eccentric targets ($e > 0.02$), the results are more restrictive. C_D estimation improves training fit but leads to unreliable propagation beyond short timescales. Baseline OD without C_D estimation provides better predictions, with errors increasing from 20–30 m at training to 200–400 m at 2 days. Maintaining sub-100 m accuracy therefore requires tracking updates within 1–2 days. The apparent 7-day performance for 2680 is not representative due to limited data and favourable geometry.

5.4.2 Mission Architecture Implications

Near-circular LEO debris presents a favourable operational profile, with predictable dynamics and reduced tracking requirements. In contrast, eccentric debris requires significantly more frequent tracking updates, increasing dependence on ground station availability and reducing operational flexibility. For an architecture based on a mothership deploying multiple satellites, this imposes constraints on scheduling and may limit the number of targets that can be serviced efficiently.

Prioritising near-circular targets offers a more scalable and lower-risk operational strategy. For eccentric targets, hybrid tracking approaches combining radar and SDLR may provide a compromise, balancing update frequency and accuracy.

5.4.3 Operational Recommendations

Based on the results of this study, the following recommendations are proposed for ARMA operations:

- (1) SDLR-based orbit determination should be used in place of TLE-based predictions for ADR targeting applications.
- (2) C_D estimation should be applied for near-circular orbits ($e < 0.01$), where it improves orbit fit without compromising short-term prediction stability.
- (3) For eccentric orbits ($e > 0.02$), C_D estimation should be avoided for propagation beyond short timescales.
- (4) Tracking updates should be performed at intervals of less than 2 days for eccentric targets to maintain acceptable prediction accuracy.
- (5) Mission planning should prioritise near-circular debris where possible to reduce tracking burden.
- (6) Multi-station tracking or hybrid radar–SDLR approaches should be investigated for improving orbit determination performance for eccentric debris.

5.5 Limitations

5.5.1 Validation Data Quality

Object 2123’s validation (a single 108 s pass of 9 measurements at high elevation) is insufficient to characterise prediction accuracy or C_D estimation viability during propagation. Object 2680’s July 4 dataset (53 measurements) is more robust, but the July 9 dataset (5 measurements) introduces substantial uncertainty. The counterintuitive baseline improvement ($329.8\text{ m} \rightarrow 140.4\text{ m}$) almost certainly reflects pass geometry rather than prediction quality, but cannot be confirmed without additional data.

5.5.2 Study Constraints

Single station

All measurements originate from Izaña-2, providing a fixed viewing direction, weak for along-track estimation. Multi-station networks would improve geometry, reduce parameter correlations, and potentially enable more reliable C_D estimation for eccentric targets, as suggested by Bamann & Hugentobler.¹⁶

Short OD arc

The 4-day OD window restricts the orbital passes and atmospheric conditions sampled, meaning the estimated C_D represents a narrow view of the drag environment. Viewing geometries are also limited, exacerbating the single-station issue. Arcs of 7–10 days would improve parameter estimation.

Limited object sampling

With one object per eccentricity regime, the proposed eccentricity threshold leaves a large range entirely unsampled. Additional objects are required to properly characterise the threshold.

Atmospheric model uncertainty

NRLMSISE-00 carries an inherent density uncertainty of 15–30%,²⁸ directly limiting OD accuracy. Higher-fidelity models (HASDM , JB2008) are not publicly available, so this uncertainty cannot be reduced within this analysis.

Limited published benchmarks

Most published SDLR OD studies report training-phase RMS only and do not investigate C_D estimation, eccentricity-dependent results, or medium-term prediction degradation for uncooperative debris,^{14,16} preventing direct numerical comparison of key results. Findings are instead interpreted against theoretical OD expectations²¹ and atmospheric drag principles.²⁸ This represents a gap in the literature that this study begins to address, though the strength of the contribution is limited.

Isotropic drag model

Debris objects are treated as spheres in the drag and SRP models, ignoring actual geometry and attitude, whereas the effective drag area varies throughout the orbit. An approximate shape model could reduce systematic drag error without requiring full attitude knowledge.

Analytical approach

The batch least-squares estimator processes all measurements simultaneously, which is well-suited to offline OD but not real-time tracking, where a sequential filter (e.g. Kalman) would be more appropriate. The convergence threshold of 10^{-2} and maximum of 20 iterations were chosen pragmatically. A sensitivity analysis varying these parameters would strengthen confidence in these choices.

Measurement uncertainty

$\sigma = 1.0\text{ m}$ is assigned to all observations regardless of elevation or pass geometry. Residual plots clearly show larger errors at low elevations, particularly for Object 2680. Elevation-dependent weighting would better represent true measurement quality.

Chapter 6

Conclusion

This study evaluated SDLR-based orbit determination and prediction for two LEO debris objects with contrasting eccentricities in the ARMA context, comparing baseline (fixed C_D) and C_D estimation configurations against TLE propagation. The following conclusions are drawn:

- (1) **SDLR provides operationally significant improvement over TLE.** Single-station SDLR improves orbit knowledge by $18\times$ – $163\times$ during training and $2.6\times$ – $617\times$ during validation, achieving sub-200 m accuracy at 7 days for eccentric debris (140.4 m vs 3208.2 m for TLE).
- (2) **C_D estimation improves fit but reduces prediction robustness.** C_D estimation reduces training RMS by 63% (circular) and 35% (eccentric), but leads to severe degradation for eccentric orbit propagation at medium timescales.
- (3) **Eccentricity governs the viability of C_D estimation.** Results indicate a threshold behaviour, with C_D estimation beneficial for near-circular orbits ($e \lesssim 0.01$) but unreliable for eccentric orbits ($e \gtrsim 0.02$). This threshold requires validation across additional objects.
- (4) **C_D estimation increases along-track uncertainty.** Including C_D degrades along-track uncertainty by up to $10.7\times$, reflecting the inherent correlation between drag and along-track motion in range-only OD.

- (5) **Validation data limits conclusions for circular targets.** The single-pass validation dataset for Object 2123 is insufficient to quantify prediction degradation or define tracking requirements.

Future Work

The primary limitation of this study is the lack of multi-pass validation data for near-circular targets. Future work should prioritise extended datasets to quantify prediction degradation and tracking requirements. Use of full-rate data (FRD) and elevation-dependent weighting may improve estimation accuracy by reducing the influence of low-quality observations. Multi-station SDLR would improve geometry and reduce C_D /along-track correlation, potentially enabling reliable drag estimation for eccentric debris. Finally, the failure of constant C_D estimation for eccentric orbits motivates investigation of time-varying drag models or sequential estimation methods (e.g. extended Kalman filtering).